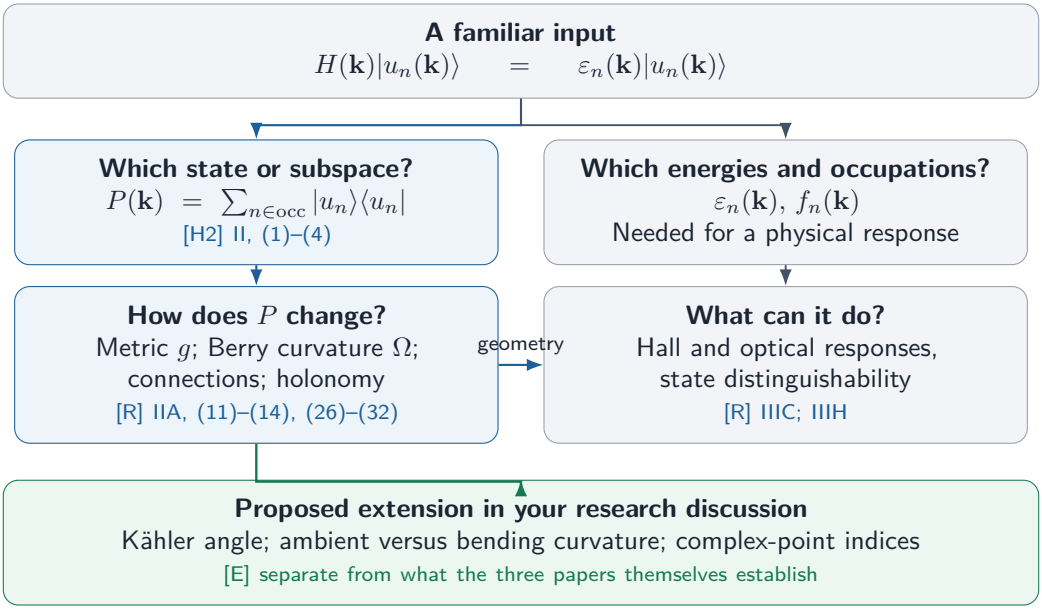
AN ILLUSTRATED READING COMPANION

From Bloch states to curved state spaces

A physics-first dictionary, relation map, and TikZ story of three papers

You already know the useful starting objects: a Hamiltonian, its eigenvectors, inner products, and projectors. Differential geometry organizes how those objects change as momentum changes. This guide builds that connection before using the advanced terminology.



Blue / [H], [R]	Established material used or reported in the cited papers. Blue does not mean that those authors originally invented the concept.
Green / [E]	The proposed multiband application supplied in this conversation. Its classical ingredients are cited separately; originality is not certified.
Gray / [Calc]	Background or a teaching illustration of standard mathematics.
Amber / conditions	Assumptions, distinctions, and questions that remain open.

Reading route. Pages 2–8 build the language; pages 9–10 tell the three-paper story; pages 11–15 unpack the extension; the final pages connect to observables, collect conventions, and give a study route.

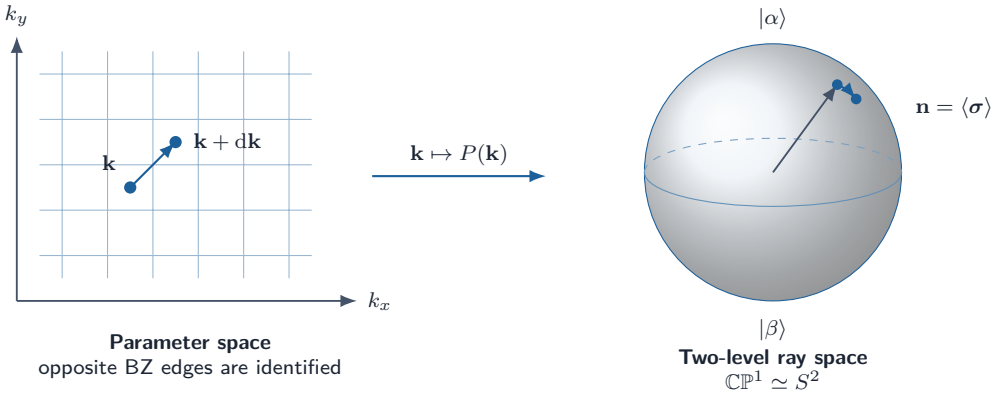
1. What is a point in state space?

Start with a two-component eigenvector, before introducing a manifold.

A point in momentum space is a value of $\mathbf{k}$. A point in pure-state space is a physical state at that value. These are different kinds of points. For two orbitals,

$$|u(\mathbf{k})\rangle = a(\mathbf{k})|\alpha\rangle + b(\mathbf{k})|\beta\rangle, \qquad |a|^2 + |b|^2 = 1.$$

Changing the *relative* amplitudes or phase can change the state. Multiplying both amplitudes by the same phase cannot change any expectation value. [R] IIA, pp. 4–6



Parameter space
opposite BZ edges are identified

Two-level ray space
$\mathbb{CP}^1 \simeq S^2$

Ray. The equivalence class of all nonzero multiples of $|u\rangle$. Normalized representatives differ by a phase:

$$|u\rangle \sim e^{i\chi}|u\rangle, \qquad P = |u\rangle\langle u| = P^2 = P^\dagger.$$

Projective space $\mathbb{CP}^{N-1}$. The set of all complex rays in $\mathbb{C}^N$. A projector is a matrix coordinate for a point in that set. [H1] II [R] IIA, (11), (38)–(43)

Why a manifold? Near a typical point, a state can be specified by a small set of smoothly varying real numbers. A sphere needs two local coordinates; an N-component normalized state modulo phase needs $2N - 2$ real coordinates. Different coordinate patches may be needed to cover the whole space.
The viewpoint. Band theory contains both an energy landscape $\varepsilon_n(\mathbf{k})$ and a map of states $P_n(\mathbf{k})$. The Bloch sphere visualizes the second object. It is not an electron trajectory in real space, a Fermi surface, or an energy surface.

The zero-metric atomic limit in the review's Fig. 1 has a momentum-independent state. A topologically trivial band with $C = 0$ can still have nonzero local metric and Berry curvature. [R] Fig. 1, p. 2; IIA

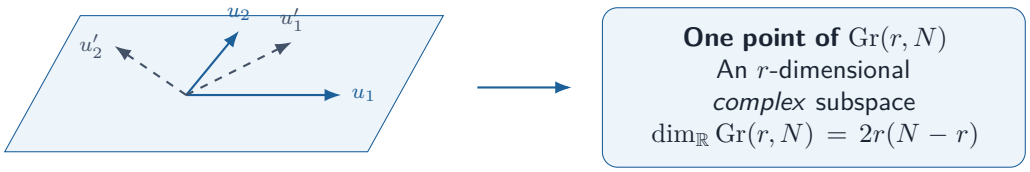
2. Several occupied bands form a moving plane

Grassmannian, frame, gauge, and bundle in linear-algebra language.

Suppose a gap separates r occupied bands from $N-r$ unoccupied bands. Collect an orthonormal occupied frame in an $N \times r$ matrix Ψ:

$$\Psi^\dagger \Psi = I_r, \qquad P = \Psi \Psi^\dagger, \qquad \Psi \mapsto \Psi U, \quad U \in U(r).$$

The last change rotates the chosen occupied basis while preserving P. It preserves subspace geometry even when individual energies differ; it need not preserve the diagonal form of the Hamiltonian. [H2] II, (1)–(4)

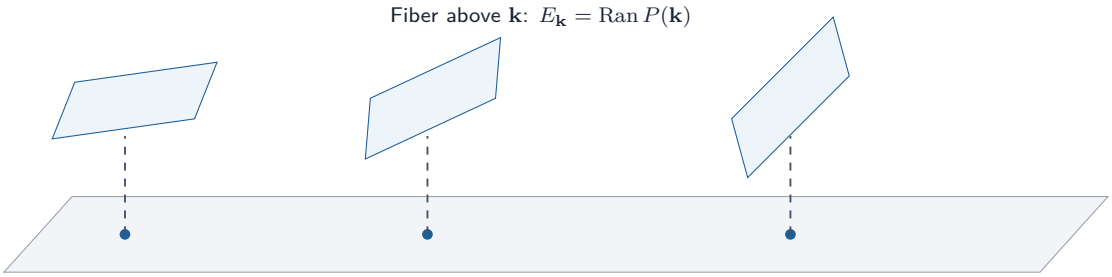


Two frames in the **same** occupied plane
$\Psi' = \Psi U \Rightarrow P' = P$

Grassmannian. The collection of all such occupied subspaces:

$$\mathrm{Gr}(r, N) = \frac{U(N)}{U(r) \times U(N-r)}.$$

The denominator removes rotations within the occupied and unoccupied sectors. Only rotations that mix the sectors change the point P. [H1] II [H2] II



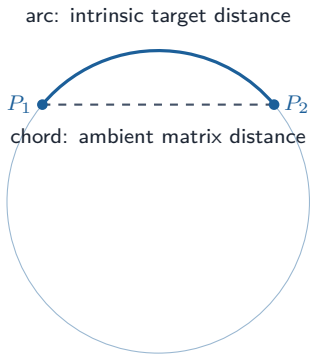
Vector bundle $E \to T^2$. Keep a copy of the occupied vector space attached to each momentum. The fibers vary smoothly while the separating gap stays open. A **local frame** is a smooth basis on a patch. A **section** chooses one vector in each fiber. A nonzero Chern number obstructs a globally smooth periodic occupied frame, while P can remain globally smooth. [H2] I–II

Keep four objects separate: base T^2; target $\mathrm{Gr}(r, N)$; image $P(T^2)$; occupied bundle E. The parameterization is regular at $\mathbf{k}$ exactly when $\mathrm{d}P_{\mathbf{k}}$ has rank two. Each sufficiently small regular source patch describes a smooth two-dimensional sheet. The complete image may have overlapping sheets or self-intersections. The full Grassmannian usually has many more dimensions. All drawn planes are real sketches of complex vector spaces.

3. Distance becomes a metric by zooming in

A finite comparison, its infinitesimal limit, and the meaning of pullback.

For two rays, define the principal angle $\vartheta = \arccos|\langle u|v\rangle|$. For two occupied subspaces, the singular values of $\Psi_1^\dagger \Psi_2$ are $\cos\vartheta_a$. These angles do not depend on the frames used. [H1] II, (4)–(5)



For one occupied band

$$D_{\mathrm{HS}} = \sqrt{1-|\langle u|v\rangle|^2} = \sin\vartheta$$
$$d_{\mathrm{FS}} = \arccos|\langle u|v\rangle| = \vartheta$$

They agree to first order for small ϑ.

[H1] II, (8) [R] IIA, (24)–(25)

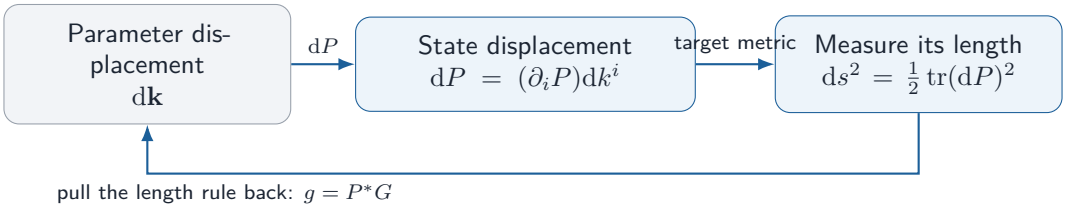
$$D^2_{\mathrm{HS}}(P_1,P_2) = \frac{1}{2}\,\mathrm{tr}(P_1-P_2)^2 = \sum_a \sin^2\vartheta_a. \qquad \text{[H1] II, (8)}$$

For $P_2 = P(\mathbf{k}+\mathrm{d}\mathbf{k})$, expand the projector:

$$D^2_{\mathrm{HS}} = g_{ij}\,\mathrm{d}k^i\mathrm{d}k^j + O(|\mathrm{d}\mathbf{k}|^3), \qquad g_{ij} = \frac{1}{2}\,\mathrm{tr}[(\partial_i P)(\partial_j P)].$$

[R] IIA, (13), (17) [H2] IV.2, (27), rescaled

Metric. A rule that turns a small displacement into a squared length. Here the length means a change of occupied subspace.



Pullback. Transfer a measurement rule from the target to the input coordinates. In the elementary analogy $\mathbf{r}(x^1,x^2)$, $g_{ij} = \partial_i\mathbf{r}\cdot\partial_j\mathbf{r}$ does exactly this. Replacing $\mathbf{r}$ by P gives the quantum version. No inverse map is needed. [MO] II and Appendix A

A shortest path in the full Grassmannian can leave the physical image $P(T^2)$. H1's disparity $\frac{2}{\pi}\sum_a \vartheta_a$ and the standard geodesic distance $\sqrt{\sum_a \vartheta_a^2}$ are different finite quantities. [H1] II, (5) [H2] IV.3–IV.4, (61)

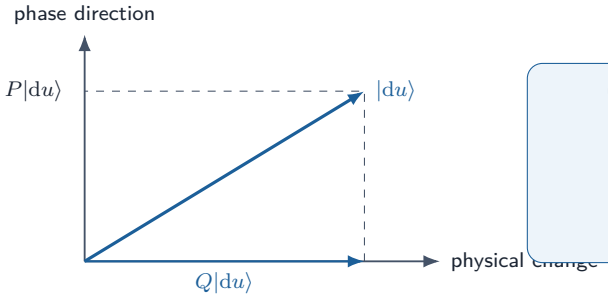
4. The QGT is a Gram matrix of state changes

This is where perturbation theory becomes geometry.

For one isolated band, split the derivative into a phase part and a physical change:

$$|\partial_i u\rangle = P|\partial_i u\rangle + Q|\partial_i u\rangle, \qquad Q = I - P, \qquad |D_i u\rangle := Q|\partial_i u\rangle.$$

$P|\partial_i u\rangle$ only changes the representative's phase; the orthogonal part changes the ray. This is the concrete meaning of a **horizontal derivative**. [R] IIA, (8)–(12), Fig. 2



Quantum geometric tensor

$$q_{ij} = \langle D_i u | D_j u\rangle$$
$$g_{ij} = \mathrm{Re}\, q_{ij}$$
$$\Omega_{ij} = -2\,\mathrm{Im}\, q_{ij}$$

[R] IIA, (12)–(14)

Two aspects of the same local data. The real inner product measures lengths and angles of state changes. The imaginary part gives the Berry phase per small oriented loop area:

$$\mathrm{d}s^2 = g_{ij}\mathrm{d}k^i\mathrm{d}k^j, \qquad \gamma_\square = \Omega_{xy}\,\delta k_x \delta k_y + \text{higher orders}.$$

[R] IIA, (4), (7), (9), (14)

A formula you can recognize from perturbation theory

Differentiating $H|u_n\rangle = \varepsilon_n|u_n\rangle$ gives, for $m \neq n$,

$$\langle u_m|\partial_i u_n\rangle = \frac{\langle u_m|\partial_i H|u_n\rangle}{\varepsilon_n - \varepsilon_m}. \quad \text{[R] IIA, (36)}$$

Therefore

$$q_{ij}^{(n)} = \sum_{m\neq n} \frac{\langle u_n|\partial_i H|u_m\rangle\langle u_m|\partial_j H|u_n\rangle}{(\varepsilon_n - \varepsilon_m)^2}. \qquad \text{[R] IIA, (37)}$$

Physics picture: momentum changes admix other bands into $|u_n\rangle$. The metric measures the size of that admixture, and the imaginary cross-correlation supplies Berry curvature.

In a fixed Bloch basis, $\partial_i H = \hbar v_i$. The same interband matrix elements also enter response theory. A small gap can enhance geometry when the numerator is nonzero; a small gap alone does not force an enhancement.

For an occupied multiplet, remove mixing inside the whole occupied subspace: $B_i = Q(\partial_i P)P$. Then $q_{ij} = \mathrm{tr}(B_i^\dagger B_j)$. The traced QGT does not contain every component of the matrix-valued non-Abelian curvature. [H2] IV.1–IV.2, (24)–(30)

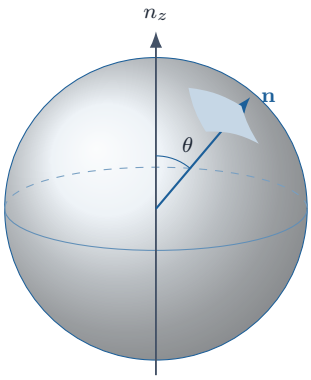
5. Work out one Bloch sphere completely

One calculation fixes metric, area, curvature, and signs.

Use a normalized state on a regular sphere-coordinate patch:

$$|u(\theta,\phi)\rangle = \begin{pmatrix} \cos(\theta/2) \\ e^{i\phi}\sin(\theta/2) \end{pmatrix}, \qquad 0 < \theta < \pi.$$

Its Bloch vector is $\mathbf{n} = (\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta)$, and $P = (I + \mathbf{n}\cdot\boldsymbol{\sigma})/2$. [R] IIA, (38)–(45)



Unit sphere drawn for direction $\mathbf{n}$

Calculated from the chosen spinor

$$q_{ab} = \frac{1}{4}\begin{pmatrix} 1 & i\sin\theta \\ -i\sin\theta & \sin^2\theta \end{pmatrix}$$

$$g_{\theta\theta} = \tfrac{1}{4}, \quad g_{\phi\phi} = \tfrac{1}{4}\sin^2\theta, \quad g_{\theta\phi} = 0$$

$$A_\phi = -\sin^2(\theta/2), \quad A_\theta = 0$$

$$\Omega_{\theta\phi} = -\tfrac{1}{2}\sin\theta$$

[R] IIA, (41)–(43) [Calc] standard-mathematics teaching derivation

Two different uses of A. $A_a = i\langle u|\partial_a u\rangle$ is the Berry connection for the chosen phase convention. The symbol dA below denotes the quantum area element. The QGT fixes g and Ω, but does not fix a gauge choice for A_a. [R] IIA, (4), (6)–(14)

$$ds^2 = \frac{1}{4}(d\theta^2 + \sin^2\theta\, d\phi^2), \qquad dA = \frac{1}{4}\sin\theta\, d\theta\, d\phi, \qquad K_G = 4.$$

The metric is that of a sphere of radius $1/2$. For this band's orientation, $\Omega_{\theta\phi}d\theta d\phi = -2\,dA$. A loop enclosing oriented solid angle $\mathcal{S}$ on the *unit* Bloch sphere has Berry phase $-\mathcal{S}/2$ modulo 2π.

An immediate distinction between two curvatures. The orthogonal state has the same metric and the opposite Berry curvature:

$$g_+ = g_-, \qquad \Omega_+ = -\Omega_-. \quad \text{[R] IIA, (41)}$$

Its intrinsic Gaussian curvature is still 4. Berry curvature therefore cannot generally be identified with the Gaussian curvature computed from g.

Pull this result back to a band. For $\theta(\mathbf{k})$ and $\phi(\mathbf{k})$, the chain rule gives $g_{ij} = \frac{1}{4}\partial_i\mathbf{n}\cdot\partial_j\mathbf{n}$ and $\Omega_{ij} = -\frac{1}{2}\mathbf{n}\cdot(\partial_i\mathbf{n}\times\partial_j\mathbf{n})$ for $P = (I + \mathbf{n}\cdot\boldsymbol{\sigma})/2$. On a rank-two patch, K_G remains 4.

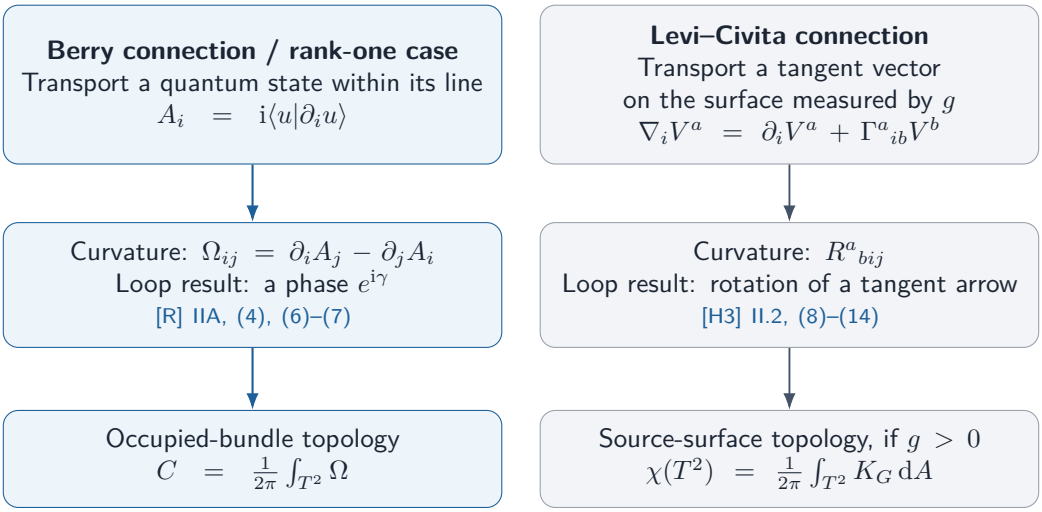
Coordinate singularity versus rank loss. The poles are singularities of (θ,ϕ) coordinates, not of the sphere. A fold of $\mathbf{k}\mapsto P(\mathbf{k})$ is different: a genuine nonzero momentum direction produces $dP = 0$. Use regular momentum coordinates to test that condition.

6. Connection means a rule for comparison

Always ask: what object is being transported?

Vectors at neighboring points may be written in different local frames. A **connection** specifies how to compare them. A **covariant derivative** corrects an ordinary derivative for the frame's change. **Holonomy** is the net transformation after a closed trip.

The **Euler characteristic** in the right-hand column counts $\chi = V - E + F$ for a triangulated surface. For a closed orientable surface with h handles, $\chi = 2 - 2h$: a sphere has 2; a torus has 0. It concerns the surface itself, whereas C below concerns its occupied bundle. [Calc] standard-mathematics teaching derivation



The familiar gauge calculation

If $|u\rangle \mapsto e^{\mathrm{i}\chi(\mathbf{k})}|u\rangle$, then

$$A_i \mapsto A_i - \partial_i\chi, \qquad \Omega_{ij} \mapsto \Omega_{ij}.$$

The gauge potential changes, while a small-loop curvature flux does not. A large-loop Berry phase is defined modulo 2π and can require multiple gauge patches. [R] IIA, (4)–(7)

What non-Abelian adds

For $r > 1$, $\mathcal{A}_i = \Psi^\dagger\partial_i\Psi$ is an anti-Hermitian matrix. Transport can rotate one occupied basis state into another. Matrix transformations depend on their order:

$$\mathcal{F}_{ij} = \partial_i\mathcal{A}_j - \partial_j\mathcal{A}_i + [\mathcal{A}_i, \mathcal{A}_j], \qquad \Omega_{ij} = \mathrm{i}\,\mathrm{tr}\,\mathcal{F}_{ij}. \quad \text{[H2] V; VI.1}$$

A Wilson loop records such a closed transport. Its determinant contains a total phase; its full matrix can contain more information. [H2] IV.5.2

The metric fixes tangent-vector transport. The Levi–Civita connection is the unique metric-preserving, torsion-free connection. Its coefficients are determined entirely by g:

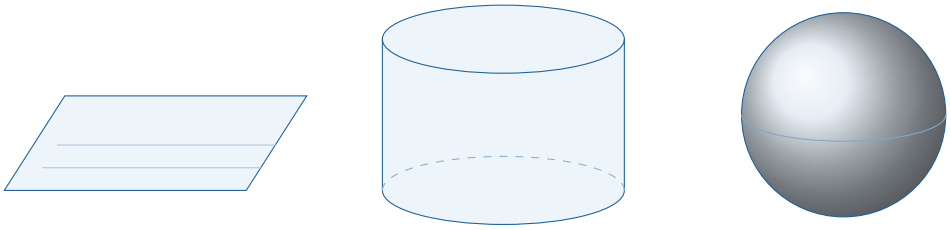
$$\Gamma^a{}_{ib} = \tfrac{1}{2}g^{ac}(\partial_i g_{bc} + \partial_b g_{ic} - \partial_c g_{ib}).$$

Christoffel symbols describe a changing tangent basis; they can be nonzero in polar coordinates on a flat plane. Curvature controls the leading transport change around an infinitesimal contractible loop. Holonomy around a noncontractible loop can be nontrivial even when curvature vanishes. [Calc] standard-mathematics teaching derivation

7. Tangent, normal, and bending

A surface can bend without becoming intrinsically curved.

Write $T_i = \partial_i P$. The tangent plane of the Bloch surface is $\mathrm{span}_{\mathbb{R}}\{T_x, T_y\}$. It is two-dimensional when these vectors are independent. An **immersion** has that independence everywhere. An **embedding** additionally avoids distinct source points meeting in the image. [H2] IV.1 [MO] II



Flat sheet
$K_G = 0$; no bending

Cylinder
$K_G = 0$; nonzero bending

Sphere of radius R
$K_G = 1/R^2$

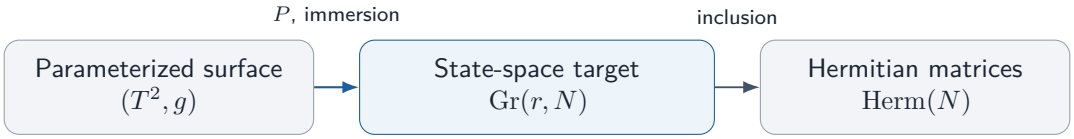
These are real-space teaching examples. A cylinder can be unrolled locally without stretching: its intrinsic metric is flat, although it bends in three dimensions. **Intrinsic** measurements use the surface metric; **extrinsic** measurements also use an ambient space. [Calc] standard-mathematics teaching derivation

First fundamental form is another name for the induced metric g. **Second fundamental form** II is the normal part of how tangent directions change:

$$\nabla_i^{\text{ambient}} T_j = \underbrace{\Gamma^k{}_{ij} T_k}_{\text{tangent part}} + \underbrace{II_{ij}}_{\text{normal part}} . \qquad \text{[AP] Appendix D, (D3)–(D5)}$$

For a surface, the Gauss equation says

$$K_G = K_{\text{ambient}}(T\Sigma) + \frac{\langle II_{xx}, II_{yy}\rangle - \|II_{xy}\|^2}{\det g}.$$



Which ambient space? The first map parameterizes the Bloch surface and may have self-intersections. The second embeds the Grassmannian in Hermitian matrices. Later, II describes each regular image sheet *inside the Grassmannian*. The Grassmannian itself is curved inside the flat matrix space. These are different second fundamental forms.

A two-band regular patch locally fills $\mathbb{CP}^1$, so its II *inside* $\mathbb{CP}^1$ vanishes. The sphere still bends inside the Hermitian matrices. Thus $K_{II} = 0$ does not mean that the sphere is flat. Also, the unoccupied Hilbert subspace $Q\mathbb{C}^N$ is not the normal space of the Bloch surface.

8. The story told by the three papers

A conceptual reading order: H1 → H2 → H3. It is not publication chronology.

H1 / Compare states at separated momenta

Occupied frames ⟶ principal angles ⟶ $D_{\mathrm{HS}}(P_1, P_2)$

Distance tables ⟶ Euclidean reconstruction by multidimensional scaling.

Question: What global shape does the sampled projector image have?

[H1] II, (4)–(8); III; Appendices A–B

zoom in; keep track of transport

H2 / Differentiate and transport the occupied subspace

$P \quad \longrightarrow \quad \partial_i P \quad \longrightarrow \quad q_{ij},\ g_{ij},\ \Omega_{ij}$

Projector paths ⟶ geodesics, Berry phases, Wilson loops, and topological invariants.

Question: Can one organize this geometry without a global frame?

[H2] IV.1–IV.5; V; VI.1

use the induced metric on a two-dimensional image

H3 / Integrate curvature, keeping track of singularities

Two-band map $T^2 \ \rightarrow \ \mathbb{CP}^1$: folds, sheet multiplicities, signed area.

Metric curvature ⟶ Gaussian-curvature integrals and singular-curve contributions.

Question: Which curvature integral counts which topology?

[H3] III–V, especially (24), (30), (34)–(38)

Open in H3's conclusion

What replaces the two-band geometric identification in multiband systems?

[H3] V, final paragraph

What the three viewpoints add

H1: global comparison. A distance table describes how far sampled subspaces are from one another. A three-dimensional projection of its reconstruction is a visualization of higher-dimensional data.

H2: local calculus and transport. Projectors provide a consistent language for changing subspaces, including multiplets for which choosing individual states is inconvenient.

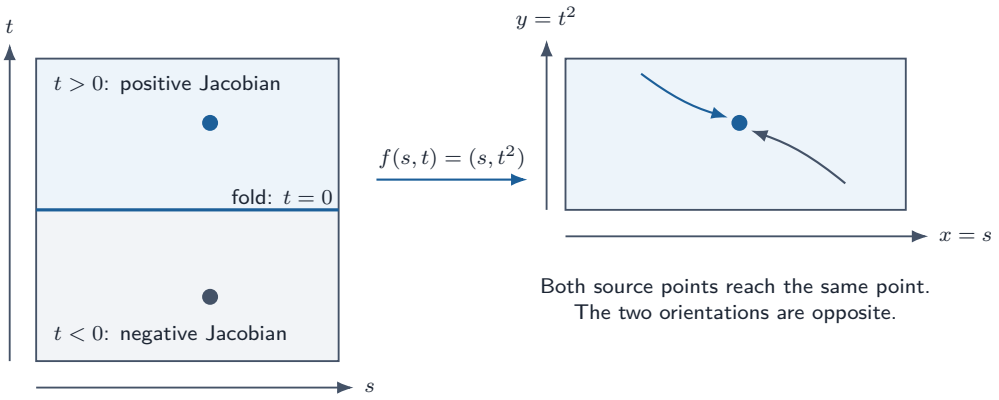
H3: careful global integration. Signed and unsigned areas count overlapping sheets differently. A smooth target does not guarantee a smooth pullback metric on the Brillouin torus.

Distances alone do not give an orientation. A plotted image also does not automatically retain how often the source covers it. H1 explicitly notes the orientation limitation. The multiband Kähler-angle and complex-point analysis on later pages is a separate extension, not a result attributed to these three papers. [H1] III.1.1, discussion after (11)

9. A fold is a failure of the map's derivative

The Hamiltonian can stay gapped while one momentum direction stops changing the state.

The simplest local fold is $f(s,t) = (s, t^2)$. Two distinct points (s,t) and $(s,-t)$ have the same image. At $t = 0$, the derivative in the t direction vanishes.



Using a flat target in this teaching example gives exactly

$$g = \begin{pmatrix} 1 & 0 \\ 0 & 4t^2 \end{pmatrix}, \qquad \mathrm{d}A = 2|t|\,\mathrm{d}s\,\mathrm{d}t, \qquad \mathrm{d}A_{\text{signed}} = 2t\,\mathrm{d}s \wedge \mathrm{d}t.$$ [Calc] standard-mathematics teaching derivation

H3's target is a sphere rather than this flat plane; the rank-loss and sheet-orientation mechanism is the same. [H3] III.1, Fig. 3

For a two-band map, in this guide's convention:

$$U = \{\mathbf{k} \in T^2 : \det g(\mathbf{k}) > 0\}.$$

$$\sqrt{\det g} = \frac{|\Omega_{xy}|}{2}, \qquad K_G = 4, \qquad \frac{1}{4\pi}\int_U K_G\,\mathrm{sgn}(\Omega_{xy})\,\mathrm{d}A = C.$$

[R] IIA, (41), with the chain rule [H3] IV, (37), converted conventions

On U, the signed curvature density is 2Ω; it extends by zero across the critical set, where $\Omega = 0$. This assigns no ordinary Gaussian curvature to the degenerate metric there. **Unsigned area** counts every sheet positively. **Signed area** counts positive and negative sheets with opposite signs. The signed formula above is a degree identity. It is not ordinary Gauss–Bonnet for a smooth torus metric.

Singular curvature. Cutting along the fold produces boundaries. Their turning supplies an additional curve contribution. H3's example uses a bulk integral plus twice the fold-curve integral in its generalized Gauss–Bonnet formula. Its unsigned bulk value divided by 2π is about 2.3779, while the fold contribution compensates it. [H3] III.3; IV, (30), (34)

A globally immersive $T^2 \to S^2$ would be a covering of the sphere, which a torus cannot be. Some rank loss is unavoidable for a smooth two-band map. Generic folds can have cusp points; other special maps have branch points. A particular four-cusp formula should not be applied to every singular map. [MO] Theorem 3 [H3] IV, (35)–(36), specific configuration

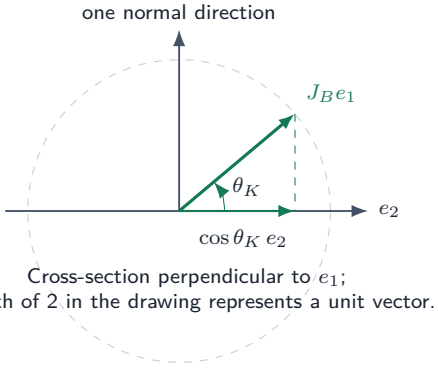
10. The missing angle in a multiband system

The next pages explain the proposed extension [E] using established geometric ingredients.

Complex structure J. On a complex vector space, multiplication by i is a real-linear operation with $J^2 = -I$. It rotates a direction through 90° in its complex line. Here it acts on *state-change vectors* B, not on the arbitrary overall phase of $|u\rangle$.

Kähler structure. The Grassmannian has a compatible metric G, complex structure J, and closed area-like two-form ω: $\omega(X, Y) = G(JX, Y)$ and $\nabla J = 0$. The induced quantum metric and Berry curvature are their pullbacks, up to the fixed factor and orientation convention. [MO] II; Appendix A

Choose e_1, e_2 as an orthonormal basis of the image tangent plane, with the orientation induced by the ordered momentum directions (∂_x, ∂_y).



Cross-section perpendicular to e_1;
a length of 2 in the drawing represents a unit vector.

At one regular momentum

$$G(e_a, e_b) = \delta_{ab}$$
$$J_B e_1 = \cos\theta_K\, e_2 + n$$
$$n \perp T\Sigma, \quad \|n\|^2 = 1 - \cos^2\theta_K$$

[MO] Appendix A, (A9)–(A14)
[E] signed-angle application

Choose the Berry-compatible complex structure J_B and define

$$\boxed{\cos\theta_K = \frac{\Omega_{xy}}{2\sqrt{\det g}}}, \qquad \Omega_{xy}\, \mathrm{d}k_x \mathrm{d}k_y = 2\cos\theta_K\, \mathrm{d}A.$$

Berry curvature is quantum area weighted by the alignment of the tangent plane with the complex structure. [E] interpretation of the Kähler pullback framework

θ_K	Tangent-plane relation	Physical geometric statement
0 or π	Complex line	$\lvert\Omega\rvert = 2\sqrt{\det g}$: determinant saturation.
$\pi/2$	Isotropic plane	$\Omega = 0$; the state can still change in two independent directions.
Intermediate	Partial alignment	$0 < \lvert\Omega\rvert < 2\sqrt{\det g}$.

The viewpoint. A sphere has no extra tangent-space direction into which $J_B e_1$ can leave the image plane. A higher-dimensional target does. This is why area and Berry curvature separate beyond two bands.

The angle requires $\det g > 0$. It describes one tangent plane, not its rate of bending. With the usual holomorphic J_0, $J_B = -J_0$ and $P^*\omega_{\mathrm{FS}} = -\Omega/2$; the sign of the cosine reverses. Absolute values and saturation do not depend on this choice. [MO] II; Appendix A

11. A wedge tests complex independence

The algebra is a Gram determinant; the new use is its band-geometric interpretation.

For a rank-one band, treat B_x and B_y as complex vectors in $\mathbb{C}^{N-1}$ and write $z = B_x^\dagger B_y$. Then

$$\det g = \|B_x\|^2\|B_y\|^2 - (\operatorname{Re} z)^2, \qquad \frac{\Omega_{xy}^2}{4} = (\operatorname{Im} z)^2.$$

The squared complex-wedge norm is $\|B_x \wedge B_y\|^2 = \|B_x\|^2\|B_y\|^2 - |z|^2$. Subtract the two expressions to obtain

$$\boxed{\det g = \frac{\Omega_{xy}^2}{4} + \|B_x \wedge B_y\|^2.}$$

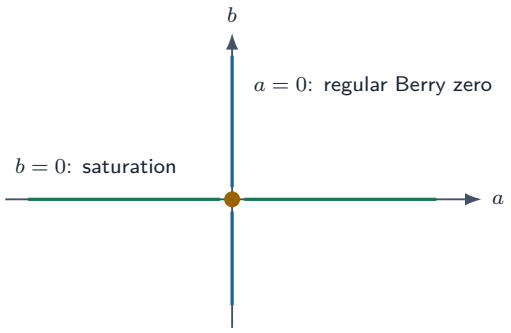
[E] application [Calc] standard-mathematics teaching derivation [MO] Appendix A, determinant inequality and equality condition

Wedge product. A generalized oriented area, with a useful algebraic test: $B_x \wedge B_y = 0$ exactly when the vectors are complex linearly dependent. It need not vanish when they are merely nonorthogonal.

A two-number example that separates all three cases

Take $B_x = (1, 0)$ and $B_y = (\mathrm{i}a, b)$, with real a, b:

$$g = \begin{pmatrix} 1 & 0 \\ 0 & a^2 + b^2 \end{pmatrix}, \qquad \Omega_{xy} = -2a, \qquad \|B_x \wedge B_y\|^2 = b^2.$$ [Calc] standard-mathematics teaching derivation



$a \neq 0,\ b = 0$: complex dependent but real independent

$a = 0,\ b \neq 0$: $\Omega = 0$ but $\det g > 0$

$a = b = 0$: metric rank loss

[E] local geometric classification

This is a plane of tangent-data parameters (a, b), not the BZ.

Why complex dependence need not collapse area. The vectors B_x and $\mathrm{i}B_x$ are dependent over $\mathbb{C}$, but independent over $\mathbb{R}$. They span a genuine real plane. The metric tests real independence; the wedge above tests complex independence.

A real physical-state example of regular Berry zero. On a local momentum patch,

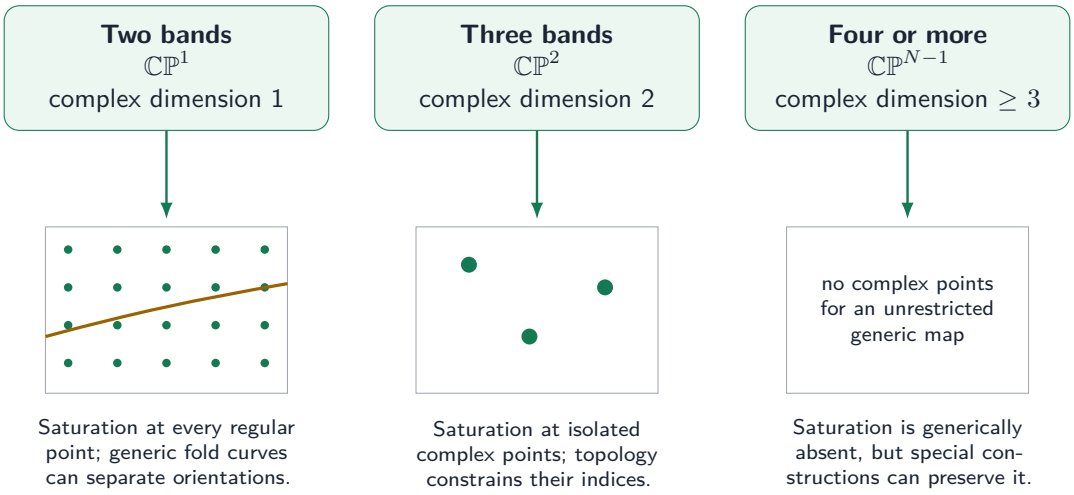
$$|u(x, y)\rangle = \frac{(1, x, y)^T}{\sqrt{1 + x^2 + y^2}} \quad \Longrightarrow \quad \Omega_{xy} = 0, \qquad \det g = (1 + x^2 + y^2)^{-3} > 0.$$

The normalized state is real and still changes in two independent directions. This local three-band example is an explicit calculation, not a periodic Chern-insulator model. [Calc] standard-mathematics teaching derivation

12. Why two, three, and more bands differ

A genericity statement for one occupied band, not a prohibition on special models.

For one occupied band, the target $\mathbb{CP}^{N-1}$ has complex dimension $N-1$. A two-dimensional Brillouin zone samples real tangent planes in that target. Requiring such a plane to be a complex line imposes more conditions as the target dimension increases. [MO] Appendix A, complex-tangent criterion [E] dimension-counting application



The rectangles are schematic BZ patches. Green dots indicate saturation; they are not numerical root positions. In the middle panel, the number drawn illustrates discreteness and is not a universal exact count.

The dimension count. If the target has complex dimension m, the complex-line locus inside the real two-plane Grassmannian has real codimension $2m-2$. A two-dimensional source generically meets:

m	1	2	≥ 3
codimension	0	2	≥ 4
saturation locus	all regular points	isolated	empty

Rank-deficient derivatives have codimension $2m-1$, explaining why generic three-band maps avoid folds. [E] transversality/dimension reasoning

Generic means stable expectation under unrestricted small smooth perturbations. It does not mean every Hamiltonian in that band count. Symmetries or a construction confined to a smaller target can change the result.

$$P(\mathbf{k}) \longmapsto P(\mathbf{k}) \oplus 0$$

adds a decoupled orbital and leaves g, Ω, and saturation unchanged. Therefore higher band number cannot, by itself, forbid ideality or complex points. For occupied rank r, use $m = r(N-r)$; the table above was stated for $r = 1$. [Calc] standard-mathematics teaching derivation

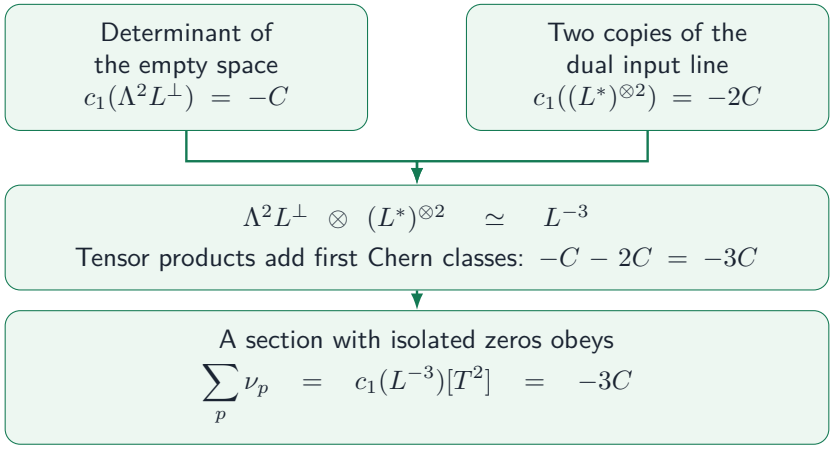
13. Why three bands carry a signed zero count

A line bundle can force a section to vanish. This is the most advanced conceptual step.

Let L be the occupied line bundle of a three-band system. The two derivatives are maps $B_i : L \to L^\perp$. Their wedge is a section of a complex line bundle:

$$W = B_x \wedge B_y \in \Gamma\left(\Lambda^2 L^\perp \otimes (L^*)^{\otimes 2}\right).$$

Here Λ^2 takes the determinant line of the two-dimensional empty space; L^* is the dual line; Γ means the collection of smooth sections. [E] bundle construction



Why the empty-space class is $-C$. The full orbital bundle is trivial in the fixed periodic basis: $L \oplus L^\perp = T^2 \times \mathbb{C}^3$. Its total first Chern class is zero. The dual changes the sign; tensor powers multiply the class. [E] elementary bundle derivation [F] (1.1), for the complex-point connection

Local winding index. Choose a gauge smooth on a disk around one zero. The phase of the local complex number representing W winds by $2\pi\nu_p$ around the disk boundary. For a simple zero, $\nu_p = \pm 1$.



Verified index pattern for one $C = -1$ example: $4 - 1 = 3 = -3C$.
These are phase-index diagrams, not locations or Berry-curvature signs.

For an immersion with simple zeros,

$$\boxed{N_{\text{complex}} = 3|C| + 2\ell, \qquad \ell = 0, 1, 2, \ldots.}$$

Topology fixes the signed sum, not the exact unsigned number. Without simple zeros, a single zero can carry higher multiplicity. [E] index consequence; five roots independently confirmed in the supplied example

The relation is a band-language instance of the classical Lai formulas. Here I_+ and I_- sum the local Bishop indices at complex points whose tangent orientation agrees or disagrees with the usual holomorphic structure J_0. They obey

$$I_+ - I_- = -3C, \qquad I_+ + I_- = \chi(T^2) + e(\nu).$$

The winding of W equals the Bishop index at a J_0-positive complex point and its negative at a J_0-negative point. Thus the subscripts on $I_\pm$ specify complex-point orientation, not the sign of the W-winding. Here $e(\nu)$ is the Euler number of the normal bundle. [L] classical index theorem [F] (1.1)

14. Where the multiband Gaussian curvature comes from

One part belongs to the Grassmannian; another comes from the Bloch surface's bending.

$$\boxed{K_G = K_{\mathrm{Gr}} + K_{II}}, \qquad K_{II} = \frac{\langle II_{xx}, II_{yy}\rangle - \|II_{xy}\|^2}{\det g}.$$

Here every inner product uses $G(A,B) = \frac{1}{2}\,\mathrm{Re}\,\mathrm{tr}(A^\dagger B)$. [AP] Appendix D, Gauss equation [E] projector specialization

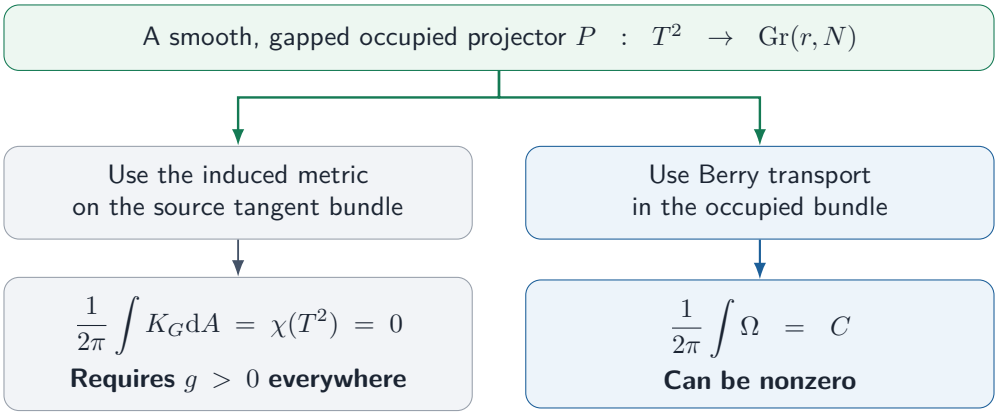
Define anti-Hermitian curvature blocks

$$\mathcal{F}_o = P[\partial_x P, \partial_y P]P, \qquad \mathcal{F}_u = Q[\partial_x P, \partial_y P]Q.$$

Their norms determine the ambient sectional curvature of the image tangent plane:

$$\boxed{K_{\mathrm{Gr}} = \frac{\|\mathcal{F}_o\|_F^2 + \|\mathcal{F}_u\|_F^2}{2\det g}.}$$ [E] Grassmann quotient curvature; Appendix A below

For rank one, this becomes $K_{\mathrm{Gr}} = 1 + 3\cos^2\theta_K$. For higher rank, the traced scalar $\Omega = \mathrm{i}\,\mathrm{tr}\,\mathcal{F}_o$ does not determine both matrix norms. The bending term can be negative. Ambient sectional curvature is nonnegative in general; in the rank-one case it lies between 1 and 4.



The viewpoint. The two integrals concern different bundles. Ordinary total Gaussian curvature measures the topology of the source surface. The Chern number measures the occupied-state bundle. This does not remove other restrictions between metric and topology, such as the volume bound $\int \mathrm{d}A \geq \pi|C|$. [MO] Theorem 2 [E] Gauss–Bonnet interpretation

Evidence in the supplied study. The three-band rank-one and four-band rank-two models have $C = -1$ and -2. Their Gauss-equation curvature agrees with a metric-only Brioschi calculation under grid refinement, and their total Gaussian-curvature integrals converge to zero. The two-band control retains constant regular curvature. [E] independent numerical checks in the earlier verification archive

A gap ensures a smooth projector; it does not prove $g > 0$. Finite grids provide evidence of immersion, not a global proof. These results address smooth Hermitian immersion geometry. A universal treatment of singular multiband maps or a new response involving K_{II} requires additional work.

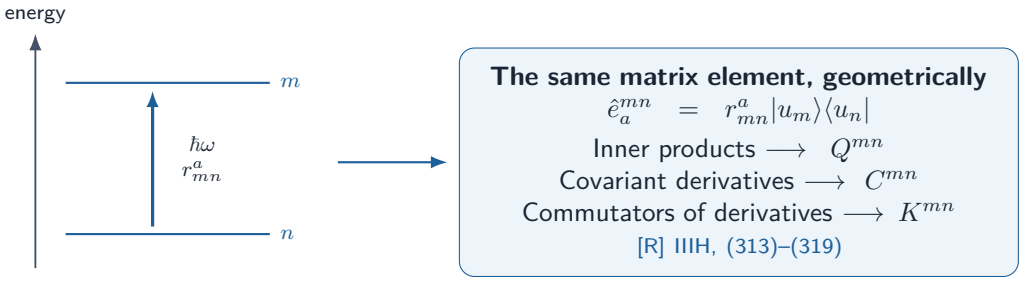
15. Where the geometry meets familiar physics

The review supplies the response context; a geometric invariant is not automatically a new observable.

For $m \neq n$, an interband dipole matrix element is

$$r^a_{mn} = \mathrm{i}\langle u_m|\partial_a u_n\rangle.$$

It describes how light polarized along a couples the two bands. The review packages it as an operator-valued tangent vector $\hat{e}^{mn}_a = r^a_{mn}|u_m\rangle\langle u_n|$. [R] IIIH, (312)–(314), pp. 27–28



Physical question	Relation and scope in the review
Intrinsic Hall response	The occupied-state Berry curvature enters the Hall conductivity. [R] IIIC, (218), p. 19
Linear optical absorption	Band-pair inner products Q^{mn} enter a resonant, occupation-weighted integral. [R] IIIH, (320), p. 28
Shift current	Covariant derivatives of transition elements enter through C^{mn}. [R] IIIH, (321), p. 28
Third-order injection	One contribution contains the optical curvature tensor K^{mn}. Other terms are present. [R] IIIH, (323), p. 28
Momentum-resolved geometry	ARPES-based reconstructions include energy-weighted quasi-QGTs and model-dependent conversion steps. [R] VD, (340)–(342), pp. 50–51

A concrete covariant derivative

A transition amplitude acquires the relative phase of its initial and final states. Its derivative must therefore include their connection difference:

$$D_c r^a_{mn} = \partial_c r^a_{mn} - \mathrm{i}(A_{m,c} - A_{n,c}) r^a_{mn}. \quad \text{[R] IIIH, (318)}$$

This is a direct physics example of correcting for a changing local frame.

The objects called curvature are different. The optical construction uses band pairs and a flag-manifold viewpoint $U(N)/U(1)^N$. Its K^{mn} is not automatically the Gaussian curvature of one band's two-dimensional quantum metric, or the scalar K_{II}. [R] Fig. 3 and (319), p. 28

What the extension would let you reconstruct. If the full $g_{ij}(\mathbf{k})$ and $\Omega_{xy}(\mathbf{k})$ are available on a region where $\det g > 0$, then $\cos\theta_K = \Omega/(2\sqrt{\det g})$ follows directly. Locating regular determinant-saturation points is then meaningful. A specific new response proportional to K_{II} is not established here. [E] operational interpretation, with the measurement assumptions retained

16. Dictionary: each word has a physics anchor

Use this as an index after reading the pictures, rather than memorizing the definitions first.

Term	Familiar object or operational picture	Page
Parameter manifold	The set of momenta or control parameters.	2
Coordinate chart	A local choice of labels, such as θ, ϕ.	2,6
Ray	All vectors representing the same pure physical state.	2
Projective space	The set of rays; two levels give the Bloch sphere.	2
Projector	A gauge-independent matrix encoding a state/subspace.	2–3
Frame / gauge	The basis or phase representatives used locally.	3,7
Grassmannian	The set of occupied r-planes in $\mathbb{C}^N$.	3
Vector bundle / fiber	The occupied vector space attached to each $\mathbf{k}$.	3
Section	A smoothly chosen vector in each fiber; zeros are allowed.	3,14
Principal angle	A basis-independent angle between two subspaces.	4
Geodesic	A locally straight path; sufficiently short segments minimize length.	4
Pullback	Measure $\mathrm{d}P$ and assign that length rule to $\mathrm{d}\mathbf{k}$.	4
Tangent vector	A first-order state/subspace change, $\partial_i P$.	5,8
Quantum metric	The real Gram matrix of physical state changes.	4–6
Berry connection	A local rule for phase or occupied-frame transport.	7
Holonomy / Wilson loop	The transformation remaining after a closed trip.	7
Covariant derivative	A derivative corrected for changing local frames.	7,16
Christoffel symbols	Coefficients describing change of a tangent basis.	7
Immersion / fold	Independent tangent directions / the simple rank-loss model $(s,t) \mapsto (s,t^2)$.	8,10
Normal / second form	Directions away from the surface / its bending.	8,15
Gaussian curvature	The intrinsic curvature of a two-dimensional metric.	6–8
Sectional curvature	Ambient curvature assigned to one tangent two-plane.	15
Complex structure	Multiplication by i on state-change directions.	11
Kähler angle	Alignment of a real tangent plane with its complex rotation.	11
Wedge / complex point	Complex independence test / saturation at an immersion.	12–14
Chern number / Euler characteristic	C concerns the occupied bundle; $\chi(\Sigma)$ concerns the source surface.	7,15

The central dependency chain

$$\text{state} \longrightarrow P \longrightarrow \mathrm{d}P \longrightarrow q = g - \frac{\mathrm{i}}{2}\Omega \longrightarrow \text{local geometry.}$$

Then split the questions: compare states; transport states; transport tangent arrows; bend a surface; integrate a curvature. Each operation uses a specific object and a specific space.

17. A convention ledger for reading the papers

The guide fixes one convention so the pictures and formulas can be checked against each other.

$$A_i = \mathrm{i}\langle u|\partial_i u\rangle, \qquad \Omega_{xy} = \mathrm{i}\,\mathrm{tr}(P[\partial_x P, \partial_y P])\,, \qquad q_{ij} = g_{ij} - \frac{\mathrm{i}}{2}\Omega_{ij}.$$

$$g_{ij} = \tfrac{1}{2}\,\mathrm{tr}(\partial_i P\,\partial_j P), \qquad \mathrm{d}A = \sqrt{\det g}\,\mathrm{d}k_x\mathrm{d}k_y, \qquad C = \tfrac{1}{2\pi}\int \Omega.$$

[R] IIA, (4), (7), (11)–(14) [H2] IV.2, (27)–(30), after metric rescaling

Quantity	This guide / standard	Huang's doubled metric
Metric	g	$g_H = 2g$
Length	$\mathrm{d}s$	$\mathrm{d}s_H = \sqrt{2}\,\mathrm{d}s$
Area in two dimensions	$\mathrm{d}A$	$\mathrm{d}A_H = 2\,\mathrm{d}A$
Gaussian curvature	K_G	$K_{G,H} = K_G/2$
Regular two-band sphere	$R = 1/2,\ K_G = 4$	$R_H = 1/\sqrt{2},\ K_{G,H} = 2$
Integrated curvature	$K_G\mathrm{d}A$	Unchanged

[H2] IV.2, (27) [H3] II.2, (2) [Calc] standard-mathematics teaching derivation

One sign difference between the papers

H2 uses $\Omega = \mathrm{i}\,\mathrm{tr}(P[\partial_x P, \partial_y P])$. H3 defines its signed density by the opposite sign in Eq. (24), then calls it Berry curvature in Eq. (38). For identical P and coordinate orientation:

$$\Omega_{\mathrm{H3}} = -\Omega_{\mathrm{H2}}, \qquad C_{\mathrm{H3}} = -C_{\mathrm{H2}}.$$

This is separate from the metric's factor of two. The guide converts both when discussing H3's signed integral. [H2] (29) [H3] (24), (38)

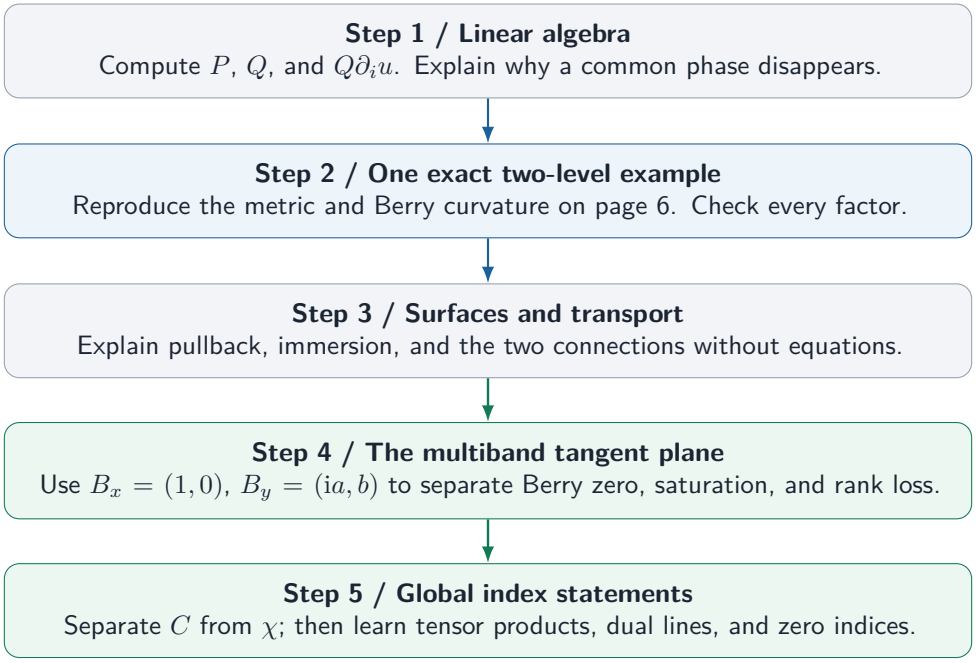
Two operations with the same letter are not necessarily the same

Symbol in this guide	Meaning
Ω_{xy}	Scalar Berry-curvature component; $\Omega = \Omega_{xy}\mathrm{d}k_x \wedge \mathrm{d}k_y$ is the two-form.
$\mathcal{F}_o, \mathcal{F}_u$	Matrix-valued anti-Hermitian curvature blocks; $\mathrm{i}\,\mathrm{tr}\,\mathcal{F}_o = \Omega_{xy}$.
$K_G,\ K_{\mathrm{Gr}}$	Source-metric Gaussian curvature; ambient Grassmann sectional curvature.
K^{mn} in the review	Optical band-pair curvature, with its own definition.
$G,\ g$	Metric on the target; metric pulled back to the Brillouin zone.

H2's $-\frac{1}{4}[[X, Y], Z]$ formula in III.2 is for the unitary group $U(N)$. Its coefficient is not the Grassmann quotient coefficient. The direct projector/Gauss derivation on page 21 avoids that substitution. Also, read the sphere radius from its metric: $g_H = \frac{1}{2}\mathrm{d}\mathbf{n}^2$ has radius $1/\sqrt{2}$, regardless of the inconsistent radius stated in H3's prose.

18. A manageable learning route

The goal is to recognize the object being measured before learning the theorem's name.



Read the supplied review selectively

Start with Fig. 1 on p. 2, then IIA on pp. 4–6. Read IIIH on pp. 27–28 after the connection pictures. Read VD on pp. 50–51 for measurement scope. The information-theory and many-body sections are valuable, but not prerequisites for this three-paper story. [R] contents and named sections

Four short self-checks

1. If $|u(\mathbf{k})\rangle = e^{\mathrm{i}\chi(\mathbf{k})}|u_0\rangle$, can $\partial_i u$ be nonzero while the metric vanishes?
2. For $B_x = (1, 0)$, $B_y = (0, 1)$, find g, Ω, and $\cos\theta_K$.
3. The two states of a two-level model have opposite Berry curvature. Must they have opposite Gaussian curvature?
4. A three-band $C = -1$ example has five simple zeros with indices $+, +, +, +, -$. Does this contradict the topological count?

Answers. (1) Yes: the derivative is pure gauge; P is constant and $Q\partial_i u = 0$. (2) $g = I_2$, $\Omega = 0$, $\cos\theta_K = 0$. (3) No: their metrics are identical, so their Gaussian curvatures agree. (4) No: the signed sum is $3 = -3C$, while the unsigned count is 5.

19. Sources and how they are used

Source codes appear beside relations and on the diagrams; all figures are newly drawn teaching schematics.

References

[R] A. Gao, N. Nagaosa, N. Ni, and S.-Y. Xu, *Quantum geometry phenomena in condensed matter systems*, Rev. Mod. Phys. **98**, 035005 (2026). doi:10.1103/f78t-ky69. The user-supplied 67-page review. Printed p. 035005-n is PDF page n.

[H1] S.-M. Huang and D. Giataganas, *Exploring Grassmann manifolds in topological systems via quantum distance.* arXiv:2412.20046v2. Finite subspace comparison and projector-distance reconstruction.

[H2] S.-M. Huang, *Quantum-State Projectors on Grassmannian: Geometry, Holonomy, and Topology.* arXiv:2608.06777v1. Projector tangents, QGT, geodesics, holonomy, and topology.

[H3] S.-M. Huang, *A Gauss-Bonnet Theorem for Quantum States: Gauss Curvature and Topology in the Projective Hilbert Space.* arXiv:2510.15760v1. Two-band singular geometry; III–V are the core of the story here.

[MO] B. Mera and T. Ozawa, *Kähler geometry and Chern insulators: Relations between topology and the quantum metric*, Phys. Rev. B **104**, 045104 (2021). arXiv:2103.11583. Pullback framework, two-band obstruction, and Appendix A's complex-tangent saturation criterion.

[AP] A. Avdoshkin and F. K. Popov, *Extrinsic geometry of quantum states*, Phys. Rev. B **107**, 245136 (2023). arXiv:2205.15353. Appendix D supplies the ambient/tangent/normal decomposition and Gauss equation.

[L] H.-F. Lai, *Characteristic classes of real manifolds immersed in complex manifolds*, Trans. Amer. Math. Soc. **172**, 1–33 (1972). doi:10.1090/S0002-9947-1972-0314066-8. Classical background to complex-point index formulas.

[F] F. Forstnerič, *Stein domains in complex surfaces.* arXiv:math/0201097v3 (2003). Equation (1.1) states the two combinations of complex-point indices for oriented immersed surfaces.

[E] The multiband research notes supplied in this conversation, together with the independent numerical verification summarized on 10 October 2026. Unpublished working analysis; see the earlier `kahler_gauss_audit_2026-10-10.zip`. Green indicates this proposed application, not a literature-priority judgment.

Source discipline. Standard facts and newly drawn pictures are not new research claims. The three-paper narrative stays separate from [E]. We use H1's SVD and Hilbert–Schmidt formulas rather than its problematic full-space trace expression (7), and do not universalize H3's specific four-cusp counting formula. The optical response map uses the review's (312)–(323); it does not infer a K_{II} response from the shared word curvature.

Appendix A. The projector formulas behind the pictures

Optional on a first reading. These identities make the proposed curvature calculation concrete.

At a point choose a basis where $P = \mathrm{diag}(I_r, 0)$. Differentiating $P^2 = P$ yields off-diagonal tangents:

$$T_i = \partial_i P = \begin{pmatrix} 0 & B_i^\dagger \\ B_i & 0 \end{pmatrix}, \qquad g_{ij} = \mathrm{Re}\,\mathrm{tr}(B_i^\dagger B_j). \quad \text{[H2] IV.1–IV.2, (19)–(30)}$$

The tangent projection onto the Grassmannian is $\Pi_{\mathrm{Gr}} A = PAQ + QAP$. Differentiating idempotency twice gives the normal part of the Grassmannian's own embedding in flat Hermitian matrices:

$$\mathcal{N}_{ij} = \begin{pmatrix} -(B_i^\dagger B_j + B_j^\dagger B_i) & 0 \\ 0 & B_i B_j^\dagger + B_j B_i^\dagger \end{pmatrix}. \quad \text{[Calc] standard-mathematics teaching derivation}$$

Insert this in the ordinary Gauss equation for that embedding, using $\langle A, B\rangle = \frac{1}{2}\,\mathrm{Re}\,\mathrm{tr}(A^\dagger B)$:

$$\begin{aligned} K_{\mathrm{Gr}} &= \frac{\langle \mathcal{N}_{xx}, \mathcal{N}_{yy}\rangle - \|\mathcal{N}_{xy}\|^2}{\det g} \\ &= \frac{\|B_x^\dagger B_y - B_y^\dagger B_x\|_F^2 + \|B_x B_y^\dagger - B_y B_x^\dagger\|_F^2}{2\det g}. \end{aligned}$$

[AP] Appendix D, Gauss identity [E] explicit block calculation

For the Bloch surface *inside* the Grassmannian, set $A_{ij} = \Pi_{\mathrm{Gr}}(\partial_i \partial_j P)$ and subtract its surface-tangent part:

$$II_{ij} = A_{ij} - T_a\, g^{ab} \langle T_b, A_{ij}\rangle.$$

Then use $K_G = K_{\mathrm{Gr}} + K_{II}$ from page 15. This uses projectors and their derivatives, with no global eigenvector gauge. [E] projector formula; requires an invertible pullback metric

What was independently compared numerically

The Gauss-equation result was compared with a Brioschi calculation made from finite differences of the metric. The relative discrepancy was the maximum absolute difference divided by the maximum absolute Brioschi curvature, on the subset $\det g > 10^{-3} \max_{\mathrm{grid}} \det g$.

Model	120^2	240^2	480^2
Three bands, rank 1	$2.88\,10^{-2}$	$7.98\,10^{-3}$	$1.86\,10^{-3}$
Four bands, rank 2	$4.25\,10^{-2}$	$1.11\,10^{-2}$	$2.94\,10^{-3}$

The decrease is consistent with second-order finite-difference convergence. At 240^2, the cutoff excludes about 0.9% and 12.5% of the respective grids. The five confirmed complex points test the rank-one three-band index sum; they do not certify that no additional canceling pairs exist. [E] earlier verification archive

What you can now read in the original notation. ∂P is a change of occupied subspace; g is its Gram matrix; Ω is a phase-curvature trace; II removes surface-tangent turning; the Gauss equation adds ambient curvature and bending. Every one of these operations has a distinct physical or geometric object.